

PRAKHAR BHANDARI

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GitHub: [Darelife](#) | Codeforces: [Darelife](#) | [Personal Website](#) | CGPA: 8.42

EDUCATION

BITS Pilani, Goa Campus

2023 – 2027

B.E. (Hons.) Computer Science · CGPA: 8.42

Relevant Coursework: Operating Systems, Computer Networks, Data Storage Technologies and Networks, Digital Design, Database Management Systems, Data Structures & Algorithms, Object Oriented Programming, Computer Programming

Technical Proficiency: C++, Python, C, Git, JJ, NextJS, PostgreSQL, Express.js, GCP, Java, Postman, MongoDB, Flask, Docker, Verilog, MySQL, GoLang, Firebase, Distributed Systems, CI/CD Pipelines, Kubernetes, Bash, Linux Fluency, Redis

WORK EXPERIENCE

Google | Software Engineering Intern

May 2026 – Aug 2026

- Interned in **Datastream (GCP)**, a serverless Change Data Capture (CDC) and replication service for real-time DB streams.
- Automated error-mapping pipelines across **5 database engines** (MySQL, PostgreSQL, MongoDB, SQL Server, & Oracle), precisely classifying **220+ customer errors** into services to prevent internal engine failures from being falsely attributed.
- Accelerated error-mapping rollouts from **1 week to automated execution**, safeguarding a tight **7-hr/month error budget** and suppressing the team's primary source of on-call pages by eliminating false-positive alerts and restoring telemetry.
- Built an **on-call telemetry dashboard** mapping SLO fast burns against pod rollouts to accelerate root-cause diagnosis.

AI Gurukul | Software Engineering Intern

May 2025 – Jul 2025

- Developed AI agents using **CrewAI** and **Gemini API** with **PostgreSQL** for **XPIA**, E-Library, and **Instagram Automater**.
- Implemented a PDF RAG pipeline combining **BM25 keyword matching** and **vector search** to reduce LLM hallucinations.
- Simulated an automated Cross Platform Injection Attack (**XPIA**) to manipulate a **Supabase Postgres DB**.

PROJECTS

PacMan: Network Packet Analyzer

Mar 2025

Full-Stack Development | github.com/Darelife/PacMan-hackenza2.0

- Won **2nd place out of 200+ teams** in a 24-hr hackathon building a web PCAP/PCAPNG network packet analyzer.
- Engineered a full-stack deep packet inspection platform using **Next.js**, **Flask**, **Scapy**, and **Pyshark** to capture live network packets, parse multi-layer protocol headers and payloads, and stream real-time telemetry metrics via WebSockets.

Codeforces Helper – Browser Extension

Dec 2024 – Apr 2025

Browser Extension Development

- Published browser extension on [Chrome Web Store](#) & [Firefox Add-ons](#) serving **2,000+ active users**.
- Engineered contest tracking, submission rating overlays, tag hiding filters, and custom problem list JSON backups, integrated with asynchronous **Codeforces REST API** sync and dynamic rating performance graphs.

Sing Assist – Real-Time Audio Analysis

Feb 2026 – Apr 2026

Systems Programming | github.com/darelife/singAssist

- Developed a low-latency vocal pitch tracking engine in **Go** utilizing **FFT** and **pYIN** algorithms to extract audio frequencies.
- Designed real-time audio signal processing with **moving-average pitch smoothing**, octave-jump suppression, and frequency tolerance matching algorithms to compare user vocal pitch against reference audio tracks and render live precision scores.

Predictive Prefetching for LLMs (Marvell Technologies)

Jan 2026 – Apr 2026

Machine Learning | github.com/Darelife/Adaptive-Predictive-Caching-for-RL-Workloads

- Designed a 3-tier predictive caching system (HBM → DRAM → NVMe) for AI/ML training, combining a **SHADE-inspired heuristic**, an **XGBoost predictor**, and a **contextual bandit** to prefetch “hot” data before the GPU needs it.
- Validated on real access traces from live **PyTorch ResNet/CIFAR-10** training runs using a custom simulator, achieving a **29.98% cache hit rate** and cutting I/O latency by **~28% (300ms → 216ms)** vs. an LRU baseline.

AWARDS & RECOGNITIONS

Expert on Codeforces (Peak Rating: 1643) | Handle: [darelife](#)

Jun 2025

- Solved **738+ competitive programming problems** and competed in **55+ contests**.

LEADERSHIP & EXPERIENCE

AlgoManiaX (Competitive Programming Club, BITS Goa) | Coordinator

Mar 2024 – Present

- Developed the official [AlgoManiaX website](#) and a [Discord Bot](#) hosted on **Google Cloud VM** for club operations.
- Created **Manim** animation videos for problem explanations; POC for BGCL 6.0 and automated social media updates.